

Merca notebook: use and definitions

Merca_Notebook.nb

English guide to the supplementary Mathematica notebook

1. Run the notebook

1. Save the .nb file in a writable folder and open it in desktop Mathematica. All definitions and example calculations are contained in this file.
2. Choose **Evaluation** > **Evaluate Notebook**. This evaluates the definitions and calculations in their original order. Using **Shift+Enter** evaluates only the selected cell or cells.
3. Let the calculation continue through all four groups. The Mellin calculation displays a progress bar from 0 to 40. With the default mesh, it processes $40 \times 2048 = 81,920$ subintervals and can take substantially longer than the finite calculations.
4. At the end, inspect the detailed report and the summary headed **Full-manuscript Mathematica verification for the manuscript**. The last cell exports the summary as `Merca_Full_Manuscript_Run_Results.pdf` and displays its path.

The PDF is written beside the saved notebook; the current working directory is used if the notebook directory is unavailable. Repeating the export updates the PDF with the same filename. Save the evaluated notebook to retain its displayed results.

The four calculation groups

Manuscript sections	Output produced in the notebook
2–3: periodic bounds	Values of the periodic constants and residue bounds, followed by their algebraic consistency checks.
3–4: finite checks	Selected denominator family, root bounds, p_0 and n_b ; the numerical root-of-unity check; and the finite test polynomials.
5–6: analytic bounds	Analytic scales and endpoint diagnostics, then the Mellin interval upper bounds and their checks.
6: final threshold	The 22-field detailed report, including the finite negative coefficients and the final sufficient threshold, followed by the summary and PDF export.

Default example and settings

The displayed example uses $(r, s, k) = (12, 1, 1)$. These values occur explicitly in the example calls; there is no separate settings panel. To change the example, update the arguments of `ClosedFormPeriodicData`, `ClosedFormEnvelopeCheck`, `MercaSystem`, and `MercaAnalyticScales` consistently, then evaluate the whole notebook again.

`MellinMajorantCheck[2048, 70]` uses 2048 subintervals per unit interval and working precision 70. The root-of-unity example also uses precision 70. `MercaReport[system1211, 200]` directly checks the 201 integer degrees from n_b through $n_b + 200$, inclusive. The variable names `system1211` and `report1211` are storage names for the example.

2. Read the detailed report

The report retains the original 22 fields and their order. The following definitions apply to the analytic cases $r \geq 4$. A field may contain a nested association; click it in the **Dataset** to view its entries.

Field	Meaning in the manuscript
Family	Selected four-part, three-part even- s , or three-part odd- s family.
EnvelopeMode	general-residue : the theoretical periodic bounds are used.
Parameters	The reduced integer triple $\{r, s, k\}$.
Parts	The parts in the selected finite denominator.
Period	Δ , Δ_1 , or Δ_2 , as appropriate.
Kappa	κ_4 , κ_e , or κ_o : the constant assigned to the periodic term.
ResidueRadius	$\mathcal{B}_4$, $\mathcal{B}_{3,s}$, or $\mathcal{B}_{3,r-s}$.
EpsilonMin	The theoretical lower bound $\kappa - \mathcal{B}$.
EpsilonMax	The theoretical upper bound $\kappa + \mathcal{B}$.
Oscillation	$2\mathcal{B}$, the difference between these bounds.
P0	A sufficient block index p_0 beyond the relevant nonnegative roots.
nb	$n_b = t_{4,p_0-1}$: the finite quotient has positive coefficients from this degree onward.
TestValuesAtP0	Exact values of all selected test polynomials at $p = p_0$.
Nb	The finite negative mass $N_b = \sum_{0 \leq j < n_b} \max\{0, -b(j)\}$.
NegativeCoefficients	Pairs $\{j, b(j)\}$ for the negative coefficients with $0 \leq j < n_b$.
AnalyticScales	The association containing X0 , Xp , Tp , Xs , and Ts .
Gamma	$\gamma = 11/4$ for four parts and $\gamma = 9/4$ for three parts.
K	$K(B, \gamma) = \lceil 796 \cdot 2^\gamma e^B \rceil$, where $B = 2\pi/\sqrt{3}r$.
NbStar	The convolution threshold N_b^* defined on the next page.
delta0	$\delta_0 = rk(k+1)/2 - sk$, the initial power of q .
N0	$N_0(r, s, k) = \delta_0 + N_b^*$, a sufficient threshold for the reduced parameters.
Verification	The direct coefficient check: CheckedRange , StrictlyPositive , and Failures .

The four similar threshold names

$$\underbrace{n_b}_{\text{nb: finite cutoff}}, \quad \underbrace{N_b}_{\text{Nb: negative mass}}, \quad \underbrace{N_b^*}_{\text{NbStar: convolution cutoff}}, \quad \underbrace{N_0}_{\text{N0: shifted cutoff}}.$$

The keys are case-sensitive: **nb**, **Nb**, and **N0** refer to different quantities.

Reference values for the default example

For $(r, s, k) = (12, 1, 1)$, independent exact calculations give

$$\begin{aligned} (\text{P0}, \text{nb}, \text{Nb}) &= (22, 11914, 1), \\ \text{NegativeCoefficients} &= \{\{49, -1\}\}, \\ \text{N0} &= 8630785233911808011. \end{aligned}$$

These are reference values for checking a fresh evaluation, rather than outputs saved from a Wolfram Kernel run.

3. Definitions and the role of the computations

Polynomial and periodic parts. The stable function names `P4[n,r,s]`, `PEven[n,r,s]`, and `POdd[n,r,s]` implement $P_3(n)$, $P_{2,s}(n)$, and $P_{2,r-s}(n)$, respectively. The names identify the denominator family. In each case,

$$p_\ell(n) = P(n) + C(n), \quad \kappa - \mathcal{B} \leq C(n) \leq \kappa + \mathcal{B}.$$

`EpsilonMin` and `EpsilonMax` are these theoretical bounds. The notebook does not enumerate a full period to find the exact extrema C^d, C^u . Accordingly, its sufficient thresholds can be larger than those obtained by inserting the exact extrema into the manuscript's tests.

Finite polynomial tests. `LowerExpressions` constructs the four lower polynomials for Cases 0–3 in Sections 3 and 4. Its internal keys `L0`–`L3` correspond to $C_{j,d}$, $D_{j,d}$, or $E_{j,d}$ in the selected family. The four-part routine returns nine endpoint or derivative tests. The three-part routine retains six tests: the five tests needed in the manuscript and the additional left-endpoint test `L1left`. That extra test is redundant when the global vertex test holds. The interpolation and root calculations use exact arithmetic.

Analytic scales. For $M \geq 4$, the fields in `AnalyticScales` are

$$\begin{aligned} X_0 &= 110M^7, \\ X_p &= \max\{X_0, 1.2 \times 10^7 M\}, & T_p &= \left\lceil \frac{10X_p^2}{3M} \right\rceil, \\ X_s &= \max\{X_0, 2 \times 10^{15} M\}, & T_s &= \left\lceil \frac{10X_s^2}{3M} \right\rceil. \end{aligned}$$

Their keys are `X0`, `Xp`, `Tp`, `Xs`, and `Ts`. `MinorRemainder` implements the detailed bound in Section 5, including the terms $(7/3)M + 3 \log M + 9$. The manuscript then bounds it by the compact majorant $30M^{3/2}\sqrt{X} + 2\sqrt{X} \log X + 2M^2$. `RootScaleEndpointChecks` numerically checks the detailed bound at the specified endpoint moduli; the uniform argument is given in the manuscript.

Convolution and the final threshold. For the selected family, $b(j)$ is the coefficient of the finite quotient and $M = r$. The report uses

$$N_b^* = \max\{16, 4n_b^2, 2T_p(r), (K(B, \gamma)N_b + 1)^2\}, \quad N_0 = \delta_0 + N_b^*.$$

For general parameters, set $d = \gcd(R, S)$, $r = R/d$, and $s = S/d$. `GeneralThreshold[R,S,k]` returns $dN_0(r, s, k)$ for $r \geq 4$. For reduced $(r, s) = (2, 1)$ or $(3, 1)$ it returns 0, using the nonnegativity results cited or proved in the manuscript. These cases do not require the analytic scales above.

Mellin interval bounds and tails. `MellinMajorantCheck` computes interval upper sums on $[0, 40]$. The two fields `HurwitzTailUpper` and `KTailUpper` contain the fixed analytic bound $1/100$ proved in Section 5. They are inputs to the certificate, rather than separately computed tail integrals. `AnalyticTailChecks` checks those stored inputs against $1/100$; `AllSafetyChecks` combines the compact interval checks with these inputs.

4. Useful calls and result checks

After the notebook definitions have been evaluated, the following calls can be used individually. Replace the example integers consistently when needed.

Call	Result
<code>ClosedFormPeriodicData[12,1]</code>	Values of κ and $\mathcal{B}$ for the three families.
<code>ClosedFormEnvelopeCheck[12,1]</code>	Exact consistency checks for those formulas.
<code>MercaSystem[12,1,1]</code>	Selected finite system, its polynomial tests, and its cutoff.
<code>RootOfUnityIdentityCheck[system1211]</code>	Numerical comparison at degrees $0, \dots, 6$.
<code>TransformedCoefficients[system1211,100]</code>	The exact list $b(0), \dots, b(100)$.
<code>FiniteNegativeData[system1211]</code>	N_b and the negative coefficient pairs below n_b .
<code>MercaAnalyticScales[12]</code>	The five analytic scales at $M = 12$.
<code>MellinMajorantCheck[2048,70]</code>	A fresh full Mellin interval calculation.
<code>MercaReport[system1211,200]</code>	The detailed report and direct check through $n_b + 200$.
<code>NormalizedThreshold[12,1,1]</code>	The sufficient reduced threshold N_0 .
<code>GeneralThreshold[24,2,1]</code>	The sufficient threshold for $(R, S, k) = (24, 2, 1)$.

What a successful check means

- `AllChecks` in `closedFormChecks` is `True` when the exact formulas agree. All entries in the report's `TestValuesAtP0` field should be positive. These finite calculations support the corresponding polynomial argument in the manuscript.
- `Passed` in `rootOfUnityCheck` is a numerical diagnostic at the tested degrees and precision. The root-of-unity identity itself is proved in Section 2.
- In `certificate`, the field `CompactIntegralSafetyChecks` should contain three `True` entries. The field `AllSafetyChecks` should also be `True`. Interpret the analytic tail fields as described on page 3.
- `StrictlyPositive` in the report's `Verification` field should be `True`, with an empty `Failures` list. This directly checks the reported finite range; positivity at every later degree also uses the polynomial proof.

If the report is incomplete

If only a few seconds elapse and the four groups have not produced their outputs, confirm that **Evaluate Notebook** was selected. An unevaluated function call is not a computed result. If earlier calculations were interrupted, quit the local kernel from the **Evaluation** menu and evaluate the whole notebook again. Read any `Failure[...]` or Mathematica message before using subsequent output. `$Aborted` means the calculation did not finish.

The ordinary analytic example requires integers $r \geq 4$, $\gcd(r, s) = 1$, $1 \leq s < r/2$, and $k \geq 1$. For unreduced parameters use `GeneralThreshold`, whose range is $R \geq 2$, $1 \leq S \leq R/2$, $k \geq 1$. Smaller meshes or changed precision may affect the interval checks. Retain the default settings when reproducing the supplied certificate.